

AI-Driven DNA Sequence Classification

Comparative Analysis on Encoded vs Raw DNA Sequences

Group 5

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https://www.shutterstock.com/search/genetic-future?image_type=illustration



LONG ISLAND UNIVERSITY

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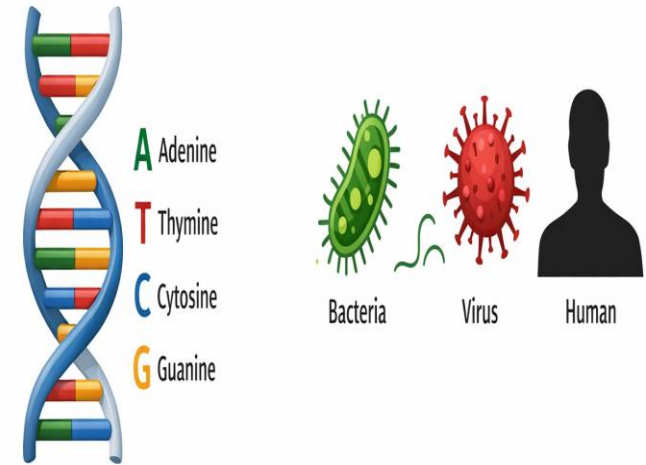
DNA sequences consist of nucleotides: A, T, C, G
Task: Splice Junction Classification

Classes:

- EI (Exon–Intron)
- IE (Intron–Exon)
- N (Non-splice)

Challenge:

- Complex biological patterns
- Large-scale genomic data



DNA Sequence: ATGCGTACGTTAGC...



GC Content



Nucleotide Counts

ATG	CGA	GCT
TAC	GCT	GAT
TCG	GAT	TCG

K-mer Features

Problem Statement

The Challenge:

- ✗ Massive growth of genomic data
- ✗ Manual DNA analysis is slow & inefficient
- ✗ Traditional ML depends on handcrafted features
- ✗ Difficulty capturing complex sequence patterns
- ✗ Learning directly from raw DNA sequences (unstructured data)

Solution:

Proposed Model: CNN - BiLSTM

CNN:

- Extracts local nucleotide motifs

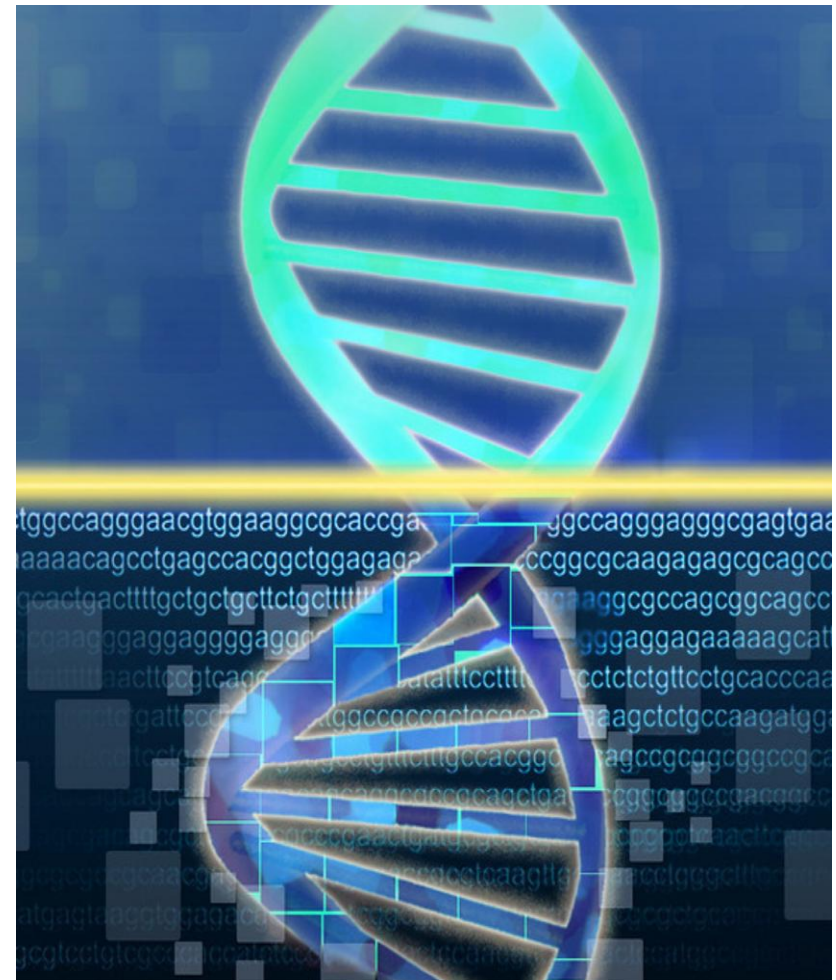
BiLSTM:

- Captures long-range dependencies

Key Advantage:

- Works on both encoded and raw DNA data

1. **Splice Site Detection (EI / IE classification)**
2. **Gene Structure Analysis**
3. **Genomic Data Processing**
4. **Disease Prediction (mutation detection)**
5. **Biomedical Research Support**
6. **Large-scale Genomic Analysis**



Research paper 1

Paper: Wavelet Neural Networks(WNN) for DNA Sequence Classification (2016)

Core Idea:

- Uses Wavelet Neural Network (WNN) for DNA classification
- Combines Genetic Algorithm (GA) + Least Trimmed Square (LTS) for optimization

Method:

- DNA → Binary encoding → Power spectrum transformation
- Wavelet-based feature extraction + K-means clustering

Key Contribution:

- Improves classification accuracy using signal-processing techniques

Limitation:

- Relies on handcrafted feature transformation
- Cannot capture deep sequential dependencies in DNA

Research paper 2

Paper: SpliceRover - Interpretable Convolutional Neural Networks for Improved Splice Site Prediction (2018)

Core Idea:

- Uses Convolutional Neural Networks (CNN) for splice site prediction
- Learns features directly from raw DNA sequences

Method:

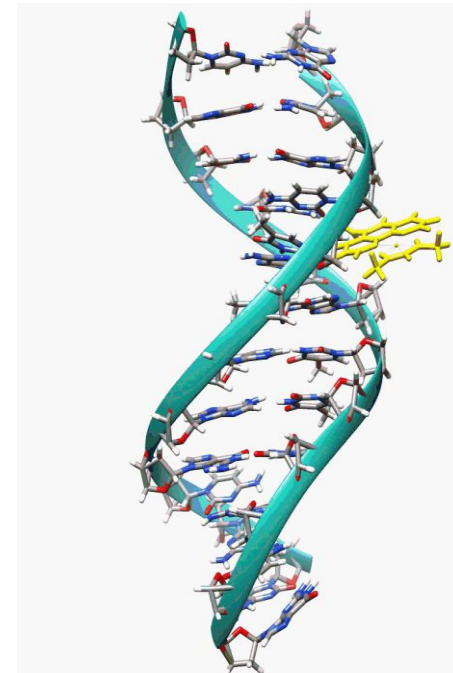
- One-hot encoding of DNA sequences
- Multi-layer CNN + max pooling + softmax classifier

Key Contribution:

- Outperforms traditional ML models (~80% improvement)
- Provides interpretability using DeepLIFT visualization

Limitation:

- Focuses only on local sequence patterns (CNN)
- Does not capture long-range dependencies



Research paper 3

Paper: A Deep Learning Approach to DNA Sequence Classification (2016)

Core Idea:

- Uses CNN with spectral (k-mer) representation of DNA
- Converts sequences into fixed-length feature vectors

Method:

- k-mer based representation (4^k features)
- CNN (LeNet-5 based) for classification

Key Contribution:

- Deep learning outperforms SVM, RF, and Naive Bayes
- Works well for short DNA fragments

Limitation:

- Ignores positional information (bag-of-words approach)
- Limited sequence dependency modeling

Aspect	Splice Rover (2018)	WNN-GA (2016)	K-mer representation (2016)
Task	Splice site prediction	DNA sequence classification (includes splice dataset)	DNA sequence classification (16S genes)
Model Used	Convolutional Neural Network (CNN)	Wavelet Neural Network + Genetic Algorithm	Convolutional Neural Network (LeNet-5 based)
Input Representation	One-hot encoded DNA sequences	Binary encoding + Fourier Transform + Power Spectrum	k-mer spectral representation
Reported Performance	Around 78-81% reduction in false discovery rate vs SVM	94% testing accuracy (splice dataset)	95-99% accuracy (full-length sequences)
Key Contribution	High-accuracy CNN with biological interpretability	Hybrid optimization improves classification robustness	Demonstrates CNN superiority over classical ML models

Research Gap

- Traditional methods → depend on feature engineering
- CNN-only models capture only local patterns
- k-mer methods → ignore sequence order
- Limited evaluation on raw DNA data
- No hybrid CNN + sequence modeling

Our contribution: Introduce a hybrid deep learning model that combines CNN and BiLSTM for both encoded and raw DNA sequences.



DATA COLLECTION

- Collect labeled DNA sequences and raw sequence of dataset
- Use benchmark dataset (e.g., Kaggle)



DATA PREPARATION

- Encode sequences (One-Hot / Numerical)
- Pad sequences to equal length
- Train / Validation split



MODEL DESIGN

- CNN → Feature Extraction
- Hybrid Architecture.
- BiLSTM → Sequence Dependencies
- Applied on : Encoded and Raw DNA Sequences Dataset



MODEL TRAINING

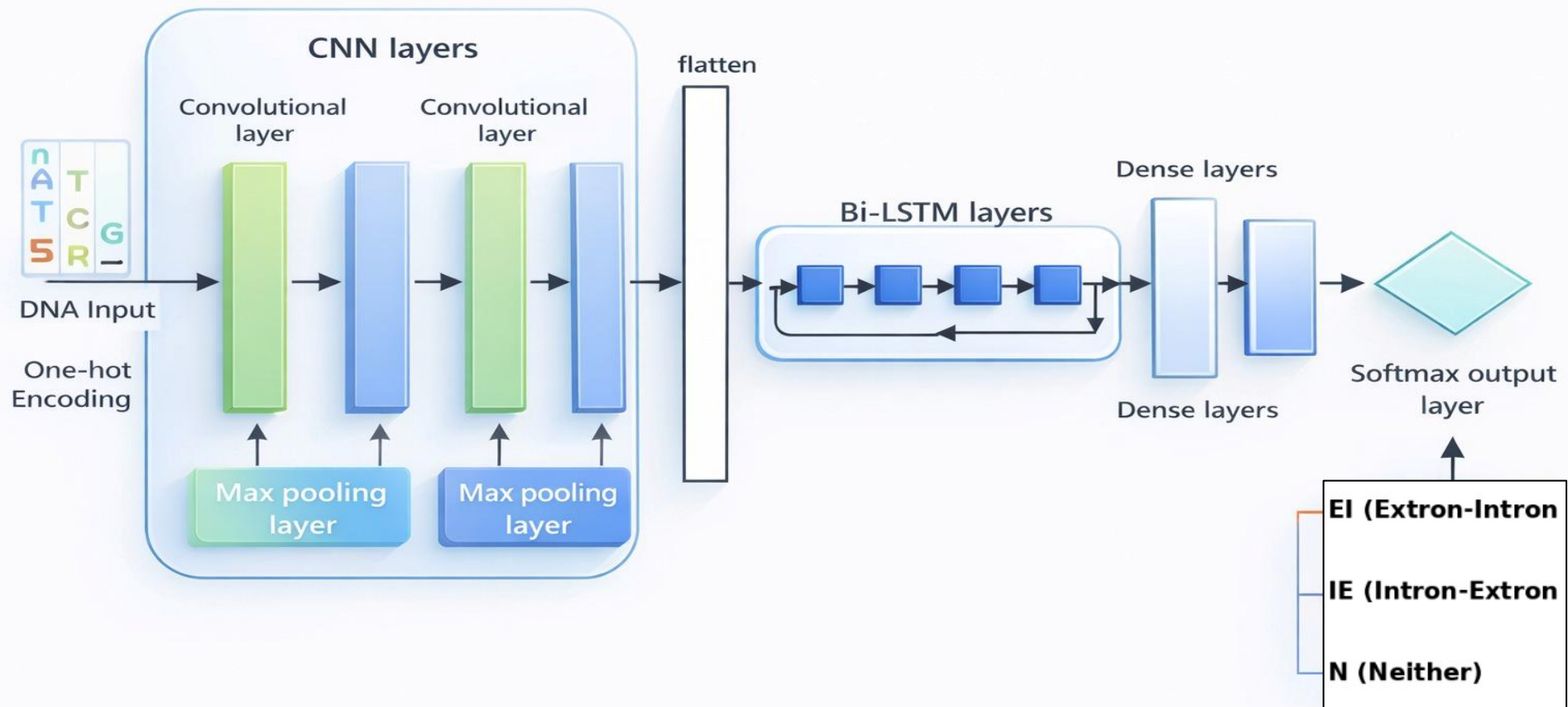
- Train on labeled data
- Loss: Cross-Entropy
- Hyperparameter tuning.



MODEL EVALUATION

- Validate on test set
- Metrics: Accuracy, F1-Score
- Error analysis.

DNA Sequence Classification Using CNN and Bi-LSTM



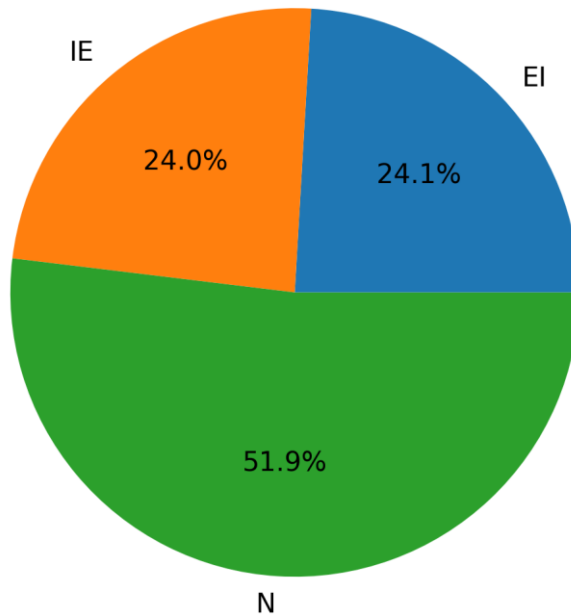
Feature	Old Dataset	New Dataset
Samples	3,190	45,619
Format	Encoded (180 features)	Raw DNA sequence
Input Shape	(60, 3)	(400, 4)
Preprocessing	Minimal	One-hot encoding
Complexity	Low	High
Realism	Moderate	High

Label	Old Dataset	New Dataset
0	N (Neither / Non-splice)	EI (Exon-Intron)
1	EI (Exon-Intron)	IE (Intron-Exon)
2	IE (Intron-Exon)	N (Non-splice)

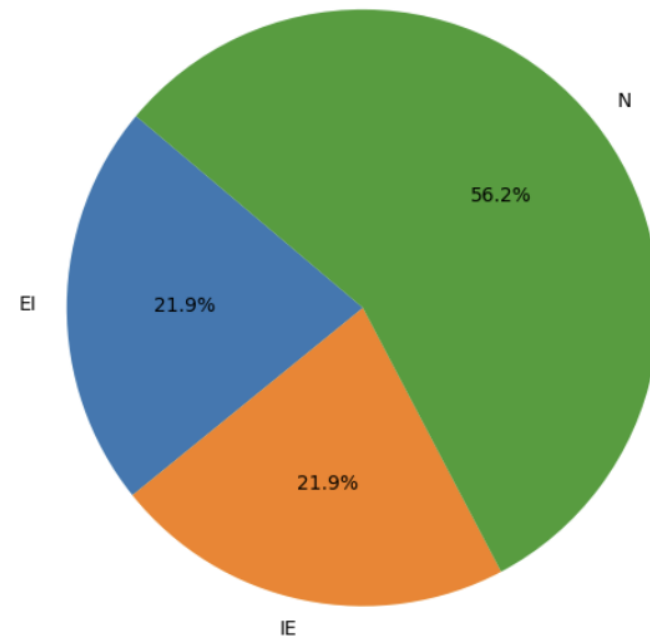
Note: Raw dataset is more complex but more realistic

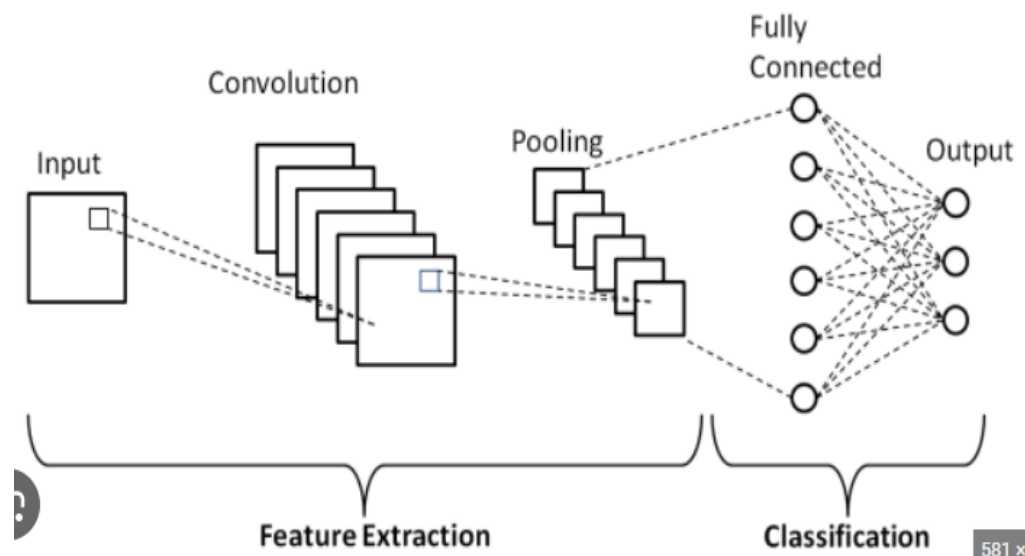
Dataset Class Distribution (Old vs New)

Class Distribution (Pie Chart)



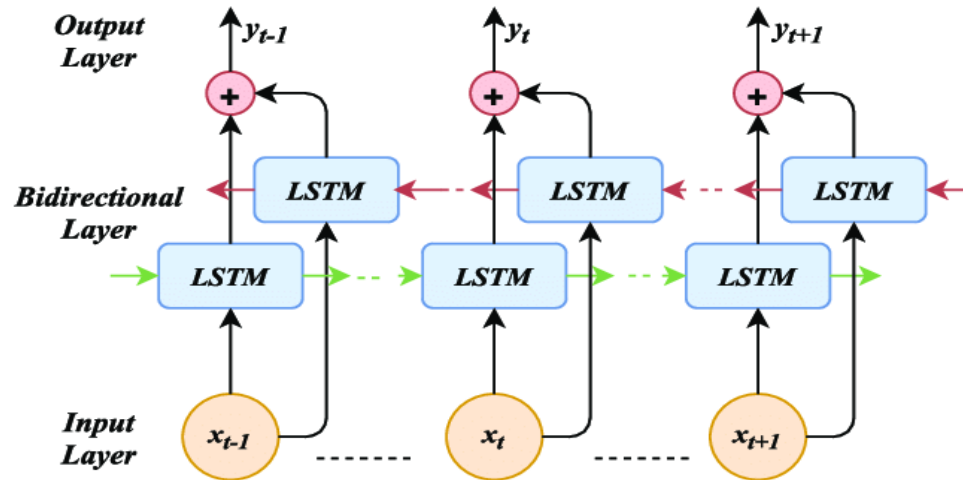
Class Distribution (EI, IE, N)





CNN Network <https://i.sstatic.net/pQHBE.png>

- Detects local nucleotide motifs
- Learns spatial patterns
- Reduces dimensionality using pooling
- Learns hierarchical feature representations automatically
- Automatically extracts features



BiLSTM Model

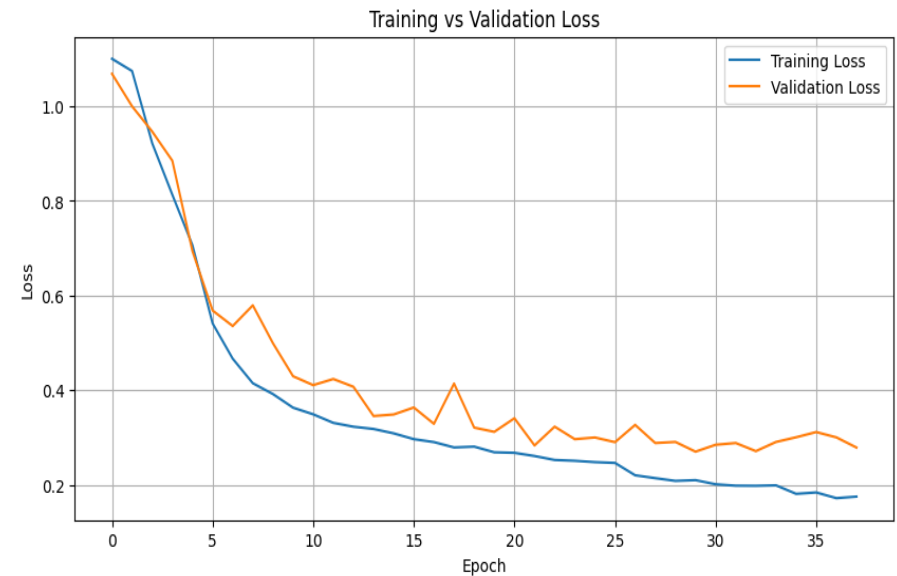
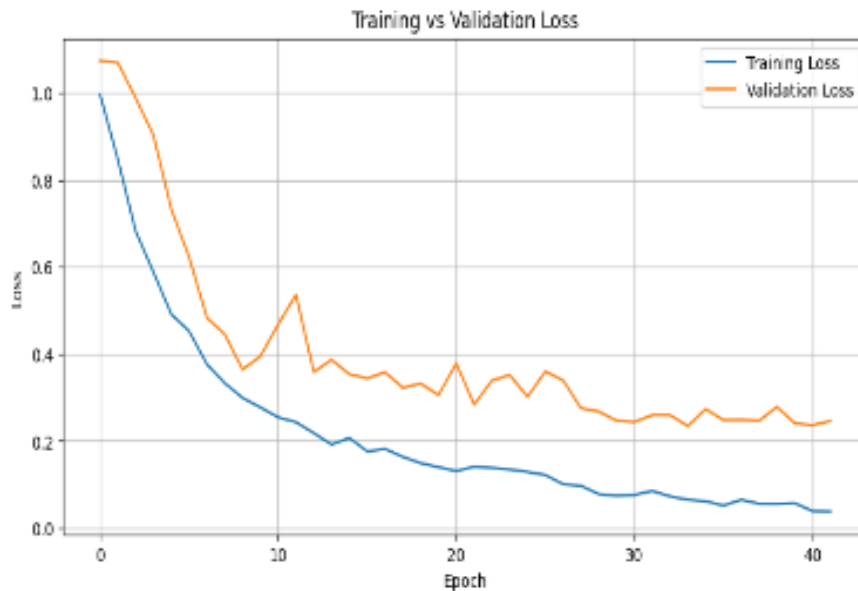
<https://medium.com/@anishnama20/understanding-bidirectional-lstm-for-sequential-data-processing-b83d6283befc>

- Captures long-range dependencies
- Processes sequences bidirectionally
- Learns contextual relationships
- Improves classification accuracy
- Enhances prediction accuracy when combined with CNN

Parameters Used:

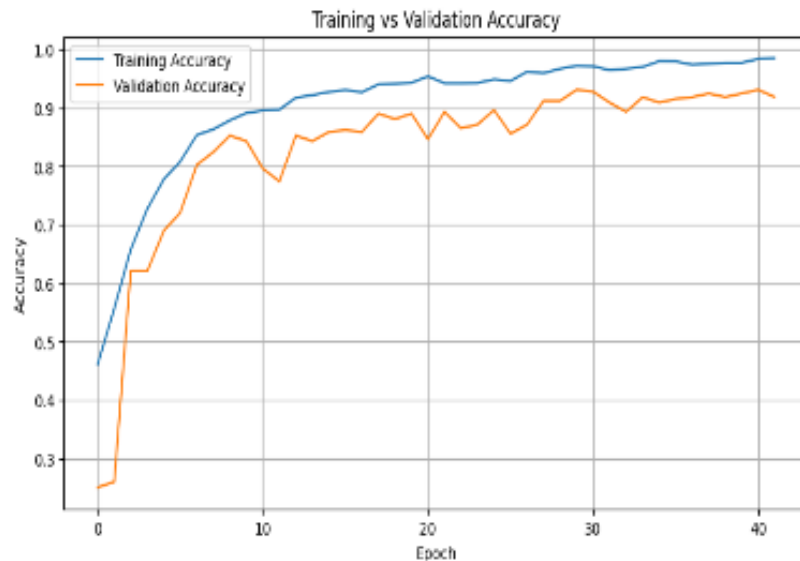
- Learning Rate: 0.001
- Batch Size: 32
- Epochs: 50
- CNN Filters: 64, 128
- BiLSTM Units: 64
- Dropout: 0.3

Training vs Validation Loss (Old & New)



Note: All the figures in comparison part → Left: Old dataset | Right: New dataset

Training vs Validation Accuracy (Old vs New)



Old Dataset:

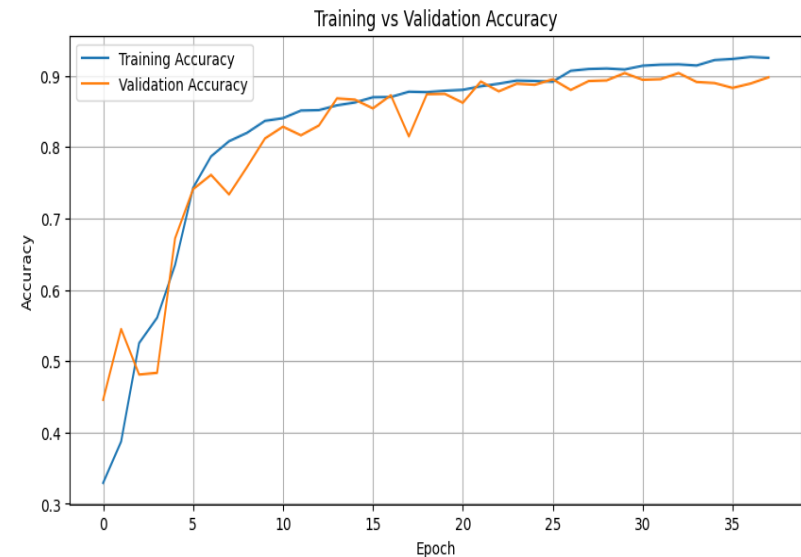
Accuracy: 94.04%

New Dataset:

Accuracy: 90.50%

Observation:

- Slight drop due to raw sequence complexity



Classification Report Of Old vs New Dataset

```
Accuracy: 0.9404388714733543
Precision: 0.9348012130006884
Recall: 0.9431457431457432
F1 Score: 0.9385975995264925
```

Classification Report:

	precision	recall	f1-score	support
0	0.93	0.99	0.96	77
1	0.92	0.91	0.92	77
2	0.96	0.93	0.94	165
accuracy			0.94	319
macro avg	0.93	0.94	0.94	319
weighted avg	0.94	0.94	0.94	319

```
Accuracy: 0.9050854888206927
Precision: 0.8842133991206605
Recall: 0.9276088992974239
F1 Score: 0.9021093718610813
```

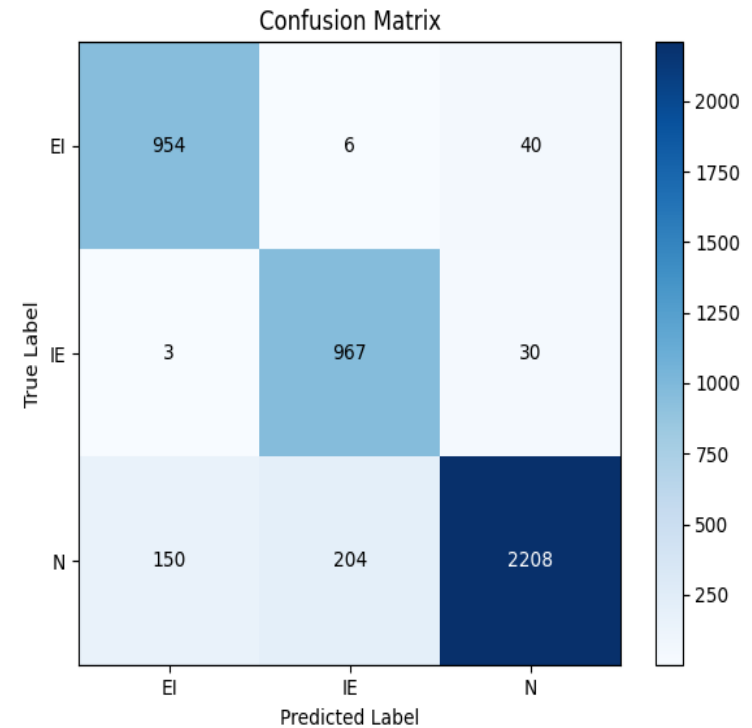
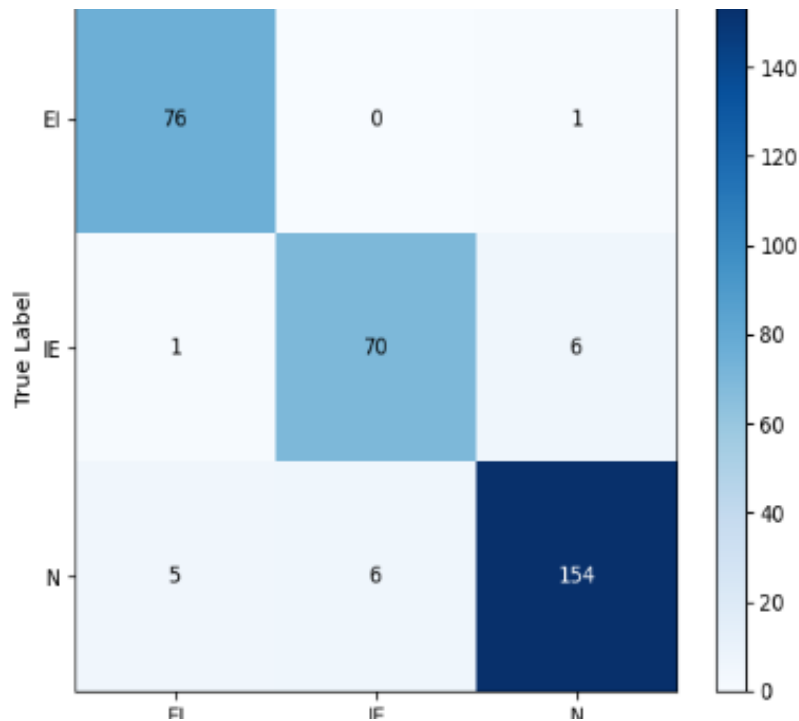
Classification Report:

	precision	recall	f1-score	support
0	0.86	0.95	0.91	1000
1	0.82	0.97	0.89	1000
2	0.97	0.86	0.91	2562
accuracy			0.91	4562
macro avg	0.88	0.93	0.90	4562
weighted avg	0.91	0.91	0.91	4562

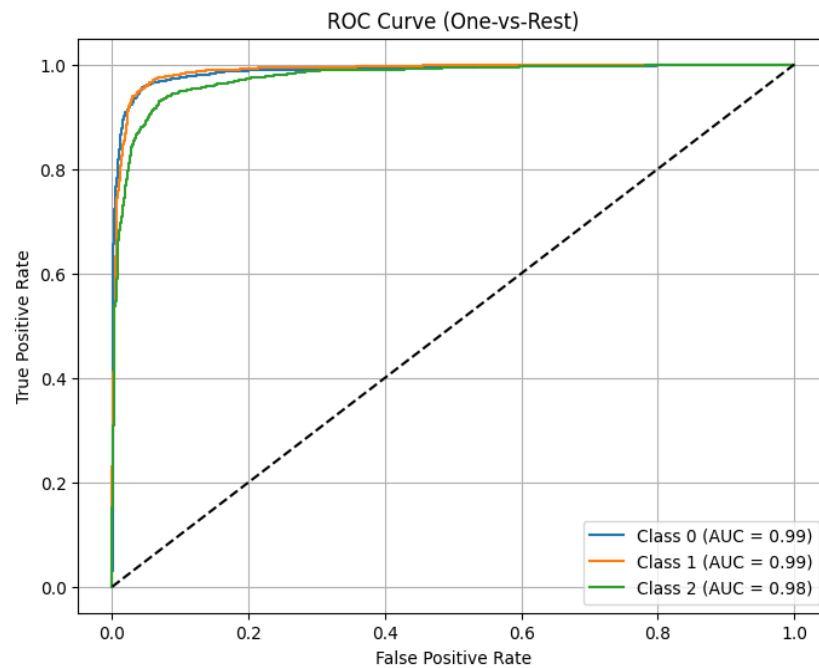
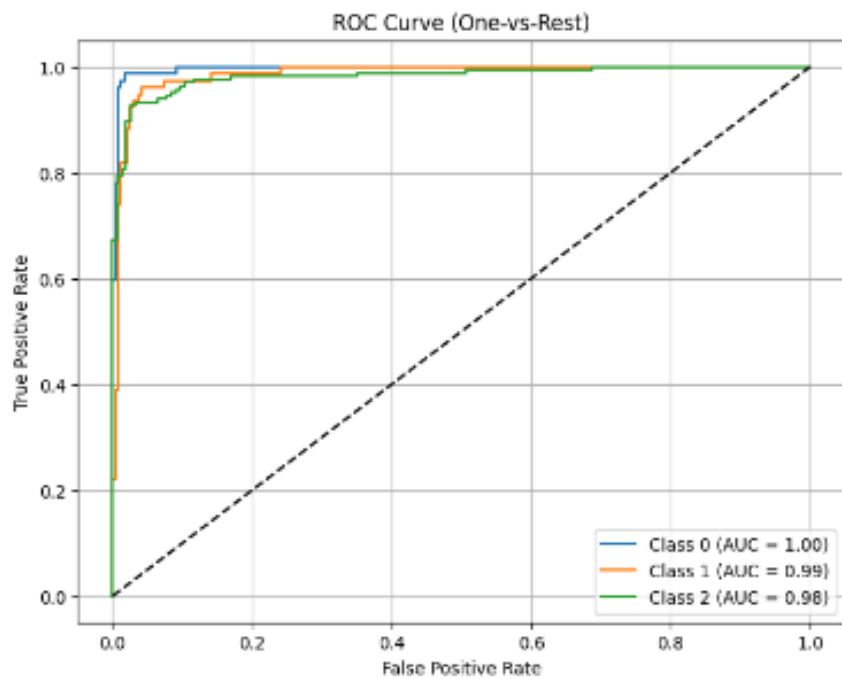
Key Insight:

Model performs better on structured data but generalizes well to raw sequences

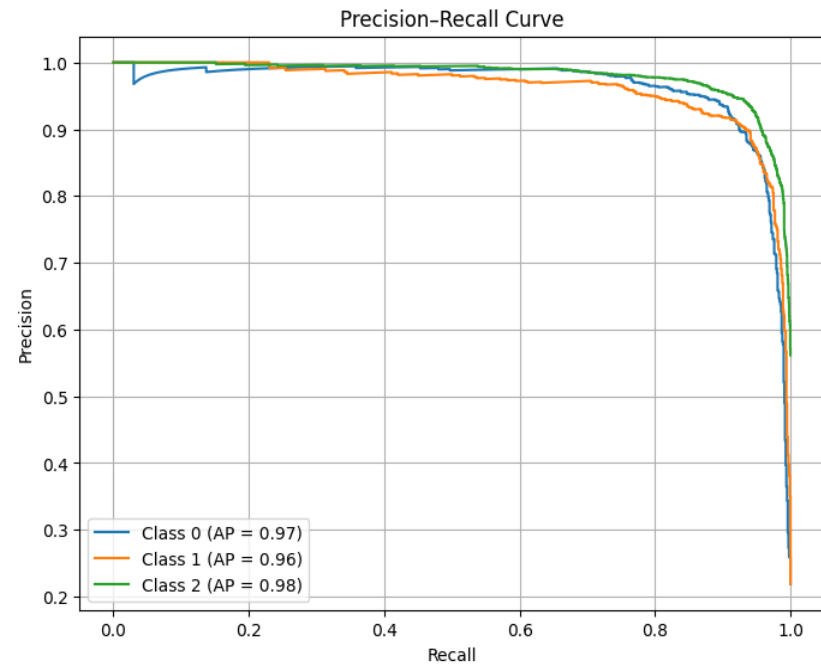
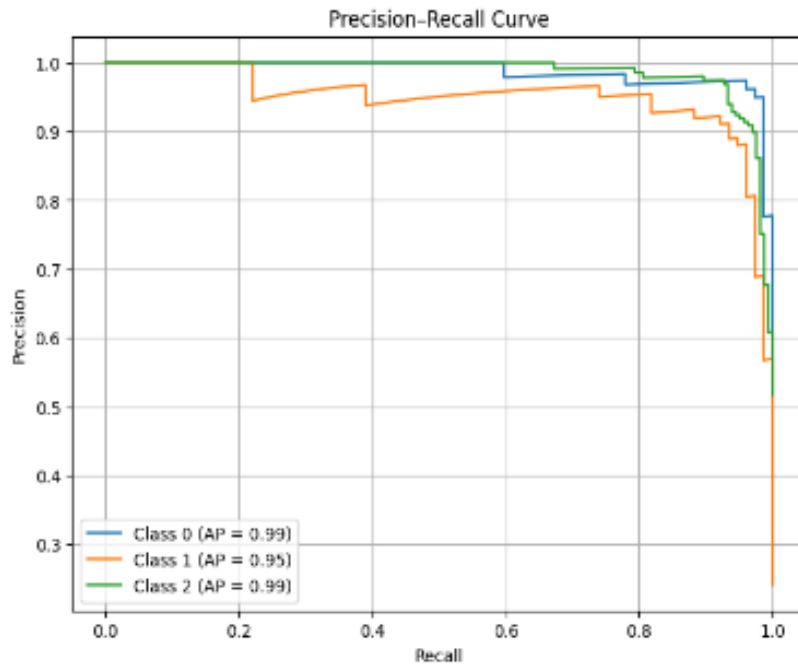
Confusion Matrix of CNN-BiLSTM Model (Old Vs New)



ROC curve showing model discrimination performance (Old vs New)



Precision-Recall curve (Old vs New)



Misclassified Samples (Old Vs New)

Total misclassified samples: 19

Index	True_Label	Predicted_Label
0	7	IE
1	30	EI
2	39	N
3	63	N
4	122	N
5	138	IE
6	143	IE
7	149	N
8	165	N
9	181	IE
10	206	N
11	211	IE
12	234	IE
13	235	N
14	257	N
15	258	N
16	269	N
17	272	IE
18	289	N

Misclassification summary:

True_Label	Predicted_Label	Count
0	EI	N
1	IE	EI
2	IE	N
3	N	EI
4	N	IE

Total misclassified samples: 433

Index	True_Label	Predicted_Label
0	6	N
1	20	N
2	31	N
3	34	N
4	48	N
5	55	N
6	59	IE
7	94	N
8	97	N
9	98	N
10	111	N
11	114	N
12	122	IE
13	124	EI
14	141	N
15	144	N
16	159	N
17	164	N
18	178	N
19	187	N

Misclassification summary:

True_Label	Predicted_Label	Count
0	EI	IE
1	EI	N
2	IE	EI
3	IE	N
4	N	EI
5	N	IE

Key Insight:

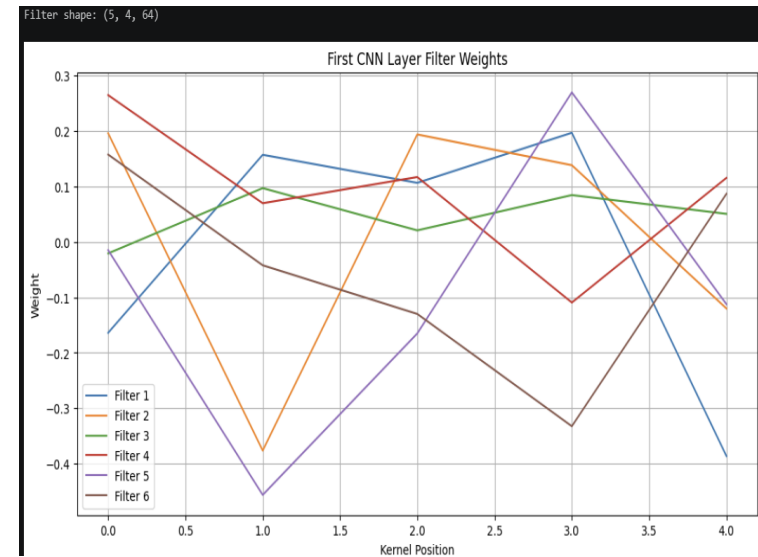
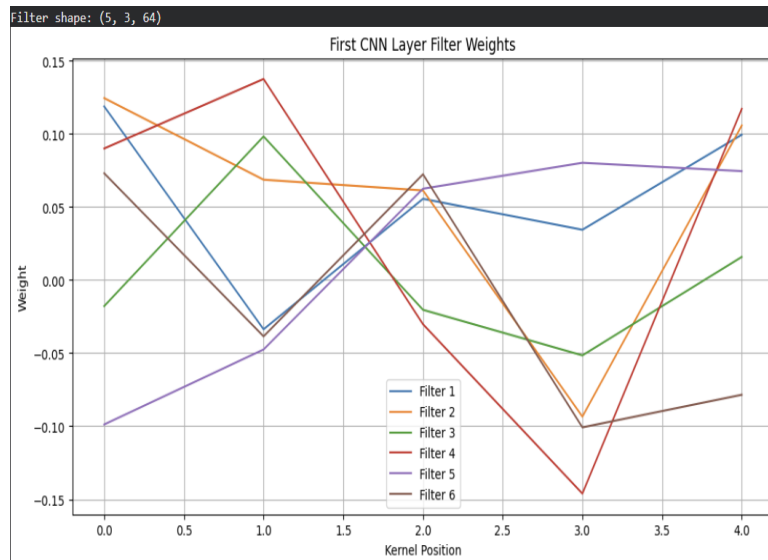
Most errors occur in Non-splice (N) class

Hyperparameters Tuning Results (Old vs New)

	BiLSTM Units	Best Validation Accuracy
0	32	0.880878
1	64	0.830721
2	128	0.890282

	BiLSTM Units	Best Validation Accuracy
0	32	0.794608
1	64	0.855984
2	128	0.865629

CNN Filter Interpretation (Old vs New)



Key Insight:

CNN filters learn biologically meaningful motifs

Metric	Old Dataset	New Dataset
Accuracy	94.04%	90.50%
Precision (Macro)	93.48%	88.40%
Recall (Macro)	94.32%	92.70%
F1 Score (Macro)	93.86%	90.20%

Model	Accuracy (Old)	Accuracy (New)
SVM	95.92%	79.15%
Random Forest	96.24%	73.69%
CNN - BiLSTM	94.04%	90.50%

OBSERVATIONS:

- High accuracy (94.04% FOR Old and 90.50% for New)
- Strong generalization
- N class highest precision
- EI / IE slight confusion
- Robust across datasets

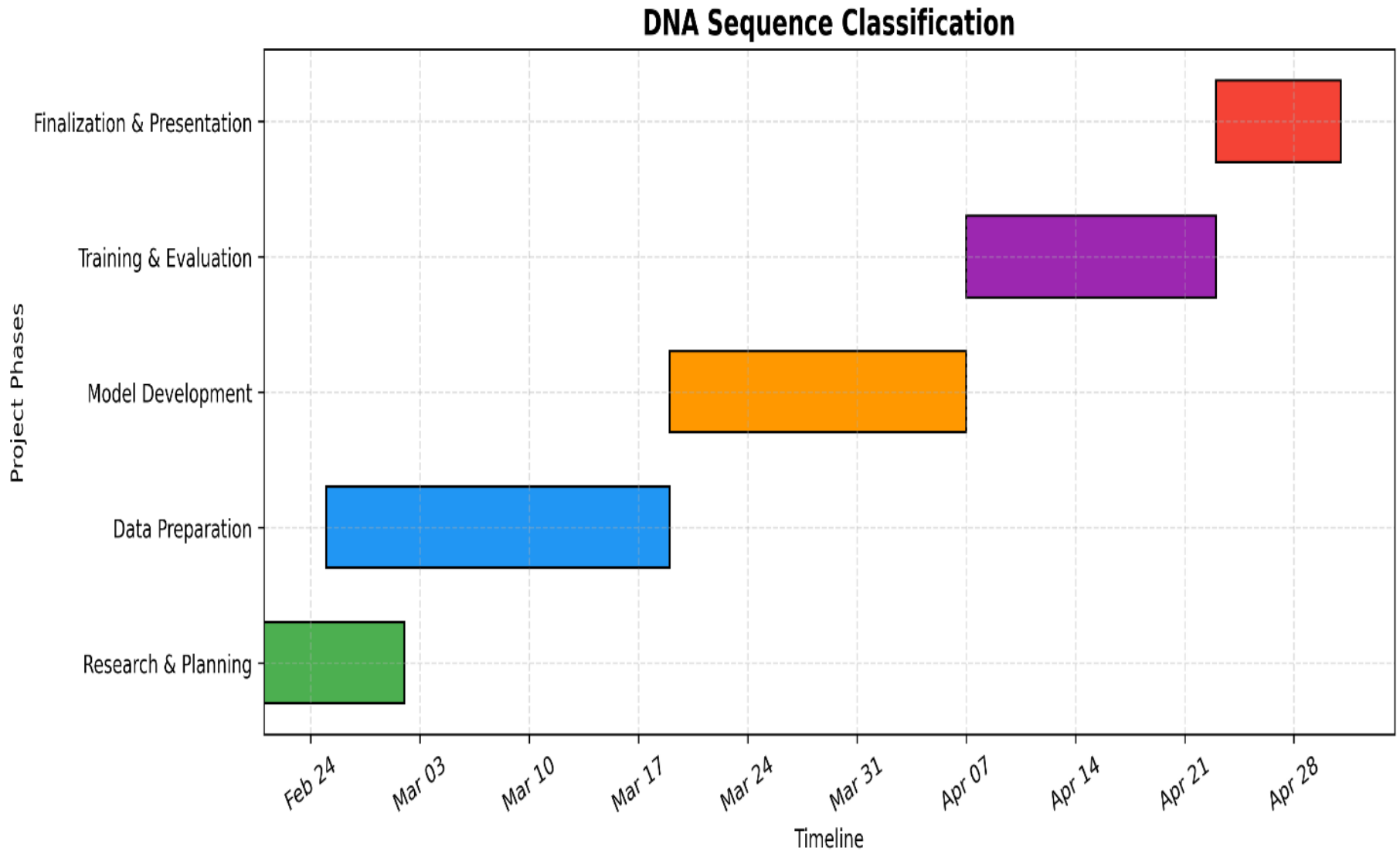
Conclusion

- CNN - BiLSTM model is effective
- Outperforms traditional models
- Captures local + sequential features
- Works on both encoded and raw data

Future Work

- Integration of attention mechanisms
- Explore Transformer-based models
- Apply Explainable AI techniques
- Use larger genomic datasets

Team Member	Primary Contributions	Responsibilities
SWAYAM	• Project Coordination & Model Design • CNN - BiLSTM Architecture Planning • Literature Review Analysis • Final Report Structuring	• Design hybrid CNN - BiLSTM model architecture • Lead weekly discussions and milestone tracking • Review and integrate research findings • Compile and finalize proposal/report
KRUPAL	• Data Acquisition & Preprocessing • Dataset Encoding & Feature Engineering • Baseline Model Implementation • Results Visualization	• Download and clean splice junction dataset • Implement one-hot encoding and data splitting • Train SVM (Support Vector Machines), and Random Forest • Generate accuracy, F1-score, and confusion matrix plots
KARTAVYA	• Hybrid Model Implementation • CNN - BiLSTM Training & Optimization • Hyperparameter Tuning • Evaluation & Analysis	• Implement CNN - BiLSTM architecture in TensorFlow/PyTorch • Train model and monitor validation performance • Perform hyperparameter tuning • Conduct comparative analysis and error evaluation



- [1] Zuallaert, J., Godin, F., Kim, M., Soete, A., Saeys, Y., & De Neve, W. (2018). SpliceRover: interpretable convolutional neural networks for improved splice site prediction. *Bioinformatics*, 34(24), 4180-4188.

- [2] Dakhli, A., & Bellil, W. (2016). Wavelet neural networks for DNA sequence classification using the genetic algorithms and the least trimmed square. *Procedia Computer Science*, 96, 418-427.

- [3] Rizzo, R., Fiannaca, A., La Rosa, M., & Urso, A. (2015, September). A deep learning approach to DNA sequence classification. In *International meeting on computational intelligence methods for bioinformatics and biostatistics* (pp. 129-140). Cham: Springer International Publishing.

- [4] Jaganathan, K., Panagiotopoulou, S. K., McRae, J. F., Darbandi, S. F., Knowles, D., Li, Y. I., ... & Farh, K. K. H. (2019). Predicting splicing from primary sequence with deep learning. *Cell*, 176(3), 535-548.



THANK YOU!!
Questions